**AP Precalculus** **Unit 3, Lesson 3.6**
Group Activity

Name: ____________________________ Partner: _______________________ Period: _________

**Context: Transformations of Sinusoidal Functions**

The general form $f(\theta) = a\sin(b(\theta + c)) + d$ encodes four independent features of a sinusoidal function. In this activity your group will decode these parameters from equations, connect them to transformations, and reason about how changes to each parameter affect the function. Choose **one tier** based on your readiness level. Everyone in your group should be prepared to explain your reasoning.

## Tier R — Reinforce

**Directions:** For each sinusoidal function below (no phase shift), complete the parameter table. Then answer the questions.

| **Function** | $a$ | $b$ | **Amplitude** | **Period** | **Midline** |
|---|---|---|---|---|---|
| $y = 3\sin(\theta) + 1$ | | | | | |
| $y = -\sin(2\theta)$ | | | | | |
| $y = \frac{1}{2}\sin(\pi\theta) - 3$ | | | | | |
| $y = 5\cos(\theta) + 2$ | | | | | |
| $y = -4\cos(3\theta) - 1$ | | | | | |

**1.** For $y = 3\sin(\theta) + 1$, what are the maximum and minimum output values?

Maximum = ____________ Minimum = ____________

**2.** For $y = -\sin(2\theta)$, how does the negative sign affect the graph compared to $y = \sin(2\theta)$?

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**3.** Use the midline and amplitude of $y = \frac{1}{2}\sin(\pi\theta) - 3$ to find the maximum and minimum values. Show your reasoning.

Maximum = midline + amplitude = ______ + ______ = _________

Minimum = midline − amplitude = ______ − ______ = _________

**4.** Write a sinusoidal function with amplitude 6, period $\pi$, midline $y = -2$, and no phase shift.

$f(\theta) =$ ___________________________________

## Tier A — Apply

**Directions:** For each function, factor the argument (if needed), then identify all four parameters. Match each function to one of the four graph cards provided by your teacher (record the letter).

**Function 1:** $f(\theta) = 2\sin\left(3\left(\theta + \frac{\pi}{6}\right)\right) - 1$

Amplitude: ________ Period: ________ Phase shift: ______________ Midline: ________ Graph card: _____

**Function 2:** $g(\theta) = -3\cos(2\theta + \pi) + 4$

First, factor the argument: $2\theta + \pi = 2(\theta + \_\_\_\_\_\_\_)$

Amplitude: ________ Period: ________ Phase shift: ______________ Midline: ________ Graph card: _____

**Function 3:** $h(\theta) = 4\sin\left(\frac{1}{2}\theta - \frac{\pi}{4}\right)$

First, factor the argument: $\frac{1}{2}\theta - \frac{\pi}{4} = \frac{1}{2}(\theta - \_\_\_\_\_\_\_)$

Amplitude: ________ Period: ________ Phase shift: ______________ Midline: ________ Graph card: _____

**Function 4:** $k(\theta) = \cos\left(2\theta - \frac{\pi}{3}\right) + 2$

Factor: ______________________

Amplitude: ________ Period: ________ Phase shift: ______________ Midline: ________ Graph card: _____

**Discussion:** Functions 2 and 4 are cosine functions. Could each be rewritten as a sine function? Explain what additional phase shift would be needed.

**Tier E — Extend**

**Directions:** Factor each argument, identify all parameters, rewrite as a sine function, and write a complete verbal transformation description in correct algebraic order.

**Function A:** $p(\theta) = -3\cos(2\theta - \pi/3) + 1$

**Step 1.** Factor: $2\theta - \pi/3 = 2(\theta - \_\_\_\_\_\_\_\_)$

**Step 2.** Parameters: amplitude ________, period ________, phase shift __________, midline ________, reflected _____.

**Step 3.** Rewrite as a sine function using $\cos\theta = \sin(\theta + \pi/2)$: $p(\theta) = -3\sin\Big(2\Big(\theta - \_\_\_\_\_ + \_\_\_\_\_\Big)\Big) + 1 = -3\sin\Big(2\Big(\theta - \_\_\_\_\_\_\_\_\Big)\Big) + 1$

**Step 4.** Verbal description (in correct transformation order — horizontal scaling/shifting before vertical scaling/shifting):

________________________________________

________________________________________

________________________________________

**Function B:** $q(\theta) = 5\sin(3\theta + \pi/2) - 2$
Complete Steps 1–4 for $q(\theta)$.

Step 1 (factor): ______________________________

Step 2 (parameters): amplitude ________, period ________, phase shift __________, midline ________.

Step 3 (sine form — already sine, no change needed). Notice: $\sin(\theta + \pi/2) = \cos\theta$. Write $q$ as a cosine function: $q(\theta) =$ __________________________

Step 4 (verbal description): ______________________________

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**Challenge:** Two sinusoidal functions $f$ and $g$ have the same amplitude and period. Must they be the same function? Explain what other information is needed to uniquely determine a sinusoidal function and how many parameters are required.

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